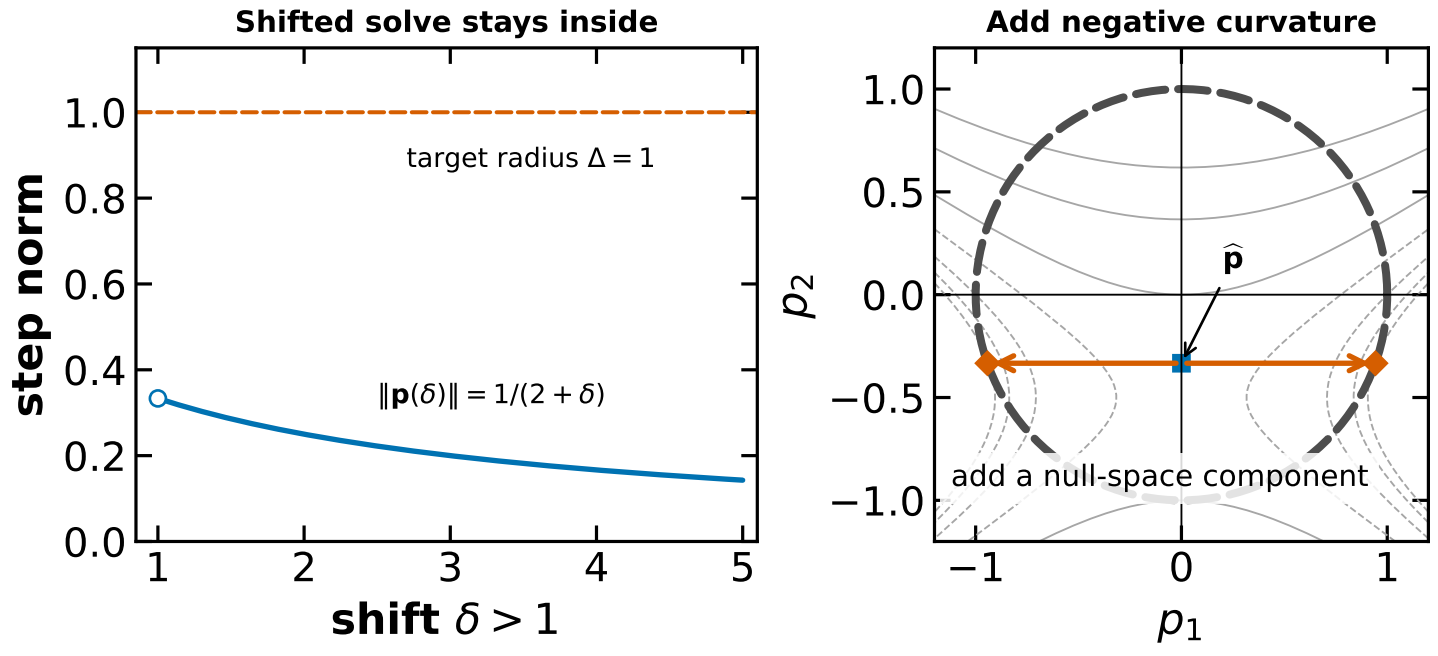
Shifted solve stays inside
step norm
target radius $\Delta = 1$
$\|\mathbf{p}(\delta)\| = 1/(2+\delta)$
shift $\delta > 1$
Add negative curvature
$\widehat{\mathbf{p}}$
add a null-space component
p_1
p_2